Set Theoretic Estimation, Recast in Cohomology

A standalone, low-jargon note

Project `thejeb`

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Abstract

This note explains, in as little jargon as possible, one idea: the gap between *local agreement* and a *global solution* in set theoretic estimation (STE) is exactly the kind of gap that cohomology was invented to measure. Because the words "presheaf" and "sheaf" scare people off, we build them from scratch with everyday examples — temperature maps, continuous functions, and one deliberately broken example that shows why the gluing axiom earns its keep — before connecting them to STE. We work one tiny example by hand and give a plain dictionary between the STE words and the sheaf words. Everything asserted as a theorem is machine-checked in the project's Lean library and cited by name; the rest is clearly flagged as intuition. No prior cohomology is assumed.

1 STE in one paragraph

Set theoretic estimation [4] models a problem as a collection of *property sets*. Each source of information — a sensor, a document, a voice — contributes a set S_i of the candidate answers consistent with that source. The *feasible set* is the intersection

$$F = \bigcap_i S_i,$$

the answers consistent with *everyone at once*. The corpus is *feasible* when $F \neq \varnothing$ and *infeasible* when $F = \varnothing$. That is the whole object of study: does a single answer satisfy every source, and if so, which ones?

2 The one idea

Checking $F \neq \varnothing$ directly means reasoning about all sources and all variables simultaneously — expensive. In practice we would rather check things *locally*: does each variable, looked at on its own, have a consistent value? Does each small group of sources agree on the piece they share?

The catch is that local agreement need not add up to a global answer. It is possible for every small piece to be internally consistent, and for every pair of overlapping pieces to agree where they overlap, and yet for *no* single global assignment to restrict to all of them at once. The pieces fit pairwise but cannot be assembled.

> **The recast.** The obstruction to assembling locally-agreeing pieces into one global solution is a *cohomology class*. When it vanishes, local agreement guarantees a global answer, and STE becomes a local computation. When it does not vanish, it measures exactly how much the problem is genuinely coupled — how far local reasoning falls short.

The mathematics that studies "local pieces and how they assemble" is sheaf theory. Its two central words, *presheaf* and *sheaf*, have fearsome reputations but modest content, and the whole recast becomes transparent once they are unpacked. So we unpack them, starting far away from STE.

3 What is a presheaf?

Forget STE for a moment. A **presheaf** is nothing more than a systematic way of talking about *data attached to regions*. It consists of two things:

1. a rule that assigns to each region U (an open set, a window, a patch — the geometry supplies these) a set $\mathcal{F}(U)$ of "data that can live on U"; and
2. for every smaller region $W \subseteq U$, a *restriction map* $\mathcal{F}(U) \to \mathcal{F}(W)$ that takes data on the big region and forgets everything outside the small one.

The restriction maps must be sane: restricting from U to U itself does nothing, and restricting in two steps (U to W, then W to W') is the same as restricting in one (U straight to W'). That is the entire definition. An element of $\mathcal{F}(U)$ is called a *section over* U, and a section over the whole space is a *global section*. Three examples make the definition feel obvious rather than abstract.

Example 1: temperature readings. Let the regions be geographic areas, and let $\mathcal{F}(U)$ be the set of possible temperature maps over the area U — one temperature per point of U. If you know the temperature everywhere in Europe, you certainly know it everywhere in France: just ignore the readings outside France. That act of ignoring *is* the restriction map. Restricting Europe to France and then France to Paris obviously gives the same answer as restricting Europe straight to Paris. Nothing deep happened; the definition of a presheaf just names the bookkeeping.

Example 2: continuous functions. Let the regions be open subsets U of the real line, and let $\mathcal{F}(U)$ be the set of continuous functions $f\colon U \to \mathbb{R}$. The restriction map is literal: given f on U and $W \subseteq U$, restrict the function, $f|_W$. A continuous function stays continuous when you shrink its domain, so the data type is preserved, and two-step restriction equals one-step restriction because both are "the same f, on a smaller set." This is the classic presheaf, and we will return to it.

Example 3: partial answers on a window (the STE one). Let the "space" be a finite set V of variables, and let a region be any subset $U \subseteq V$ — a *window* of variables we are willing to look at together. Let $\mathcal{F}(U)$ be the set of assignments of a value to each variable in U: partial answers that only speak about the window. Restriction from a big window to a smaller one forgets the values of the variables you dropped. A global section — data on the whole of V — is a complete assignment, one value for every variable. This is the presheaf STE lives in; hold the thought.

4 What makes a presheaf a sheaf?

A presheaf only records data and how to shrink it. A **sheaf** is a presheaf whose data is genuinely *local*: it can be checked, and assembled, patch by patch. Concretely, cover a region U by patches $\{U_j\}$ (so every point of U is in some U_j). The presheaf is a sheaf if two axioms hold:

Locality (separation). If two sections over U restrict to the same thing on every patch U_j, they were already equal. "Data is determined by its local values — no invisible global information."

Gluing. If you choose a section s_j on each patch, and any two choices agree on the overlap of their patches ($s_j = s_k$ on $U_j \cap U_k$), then there exists a section on all of U that restricts to each s_j. (Unique, by locality.) "Compatible local data assembles into global data."

Locality is rarely the interesting axiom; in all our examples it holds because the data literally is a function on points. Gluing is where the action lives, and the fastest way to respect it is to watch it *fail*.

A presheaf that is not a sheaf: bounded functions. On open subsets of the real line, let $\mathcal{B}(U)$ be the set of *bounded* continuous functions on U. This is a perfectly good presheaf: restricting a bounded function keeps it bounded. Now cover the whole line by the intervals $U_n = (-n, n)$ for $n = 1, 2, 3, \dots$ and put on each patch the section $s_n(x) = x$, the identity function — bounded on $(-n, n)$, since it never exceeds n there. Any two of these agree on their overlap (they are all the same formula), so the gluing axiom demands a global bounded function restricting to every s_n. The only candidate is $f(x) = x$ on all of $\mathbb{R}$ — which is *not bounded*. The compatible local family exists; the global section does not. Gluing fails, and the reason is worth stating plainly: *boundedness is not a local property*. Each patch honestly certifies its own piece, yet the certificates do not add up, because the property being certified secretly refers to the whole domain.

(A second classic failure, in one line: on a space with two separate components, "locally constant functions that agree on overlaps" can take value 0 on one component and 1 on the other — gluable as a function, but not to anything *constant*. Constancy, like boundedness, is a global property wearing local clothing.)

A presheaf that is a sheaf: continuous functions. Contrast Example 2. If continuous functions s_j on patches agree on all overlaps, define $f(x) = s_j(x)$ for any patch containing x — the agreement makes this unambiguous — and f is continuous, because continuity *is* a local property: it is checked point by point, in a small neighborhood. So continuous functions glue. The temperature presheaf glues for the same reason: a temperature map is just a value at each point, and compatible local maps trivially paste together.

> **The moral.** A sheaf is a presheaf whose data has no hidden global commitments. The gluing axiom is precisely the promise "local agreement implies a global answer" — which is the promise STE wishes it could make. Whether a given presheaf keeps that promise, and by how much it breaks it, is what cohomology measures: the failures of gluing are, in the standard framework, classes in a group called H^1, and "every compatible family glues" is the statement $H^1 = 0$.

5 The STE presheaf, and its building blocks

Now aim this machinery at STE. Fix a set V of *variables*, each variable v ranging over a value set A_v.

Global configuration. One choice of a value for every variable at once: a function f with $f(v) \in A_v$ for all v. Write the set of all of them as $\prod_v A_v$.

Constraint. A set T of allowed global configurations, $T \subseteq \prod_v A_v$. In STE, T is the feasible set F; the recast just insists on viewing F inside the full configuration space, so that restriction makes sense.

Context. A window $U \subseteq V$, as in Example 3. A small context is cheap; the full set V is the expensive global view.

Cover. A collection of contexts $\{U_j\}$ that together touch every variable. This is how we chop the global problem into local windows.

Local section. Here is where the constraint enters. On each window U_j, keep only the partial assignments that *could* be extended to a full configuration in T — "a local piece that is not obviously doomed." These extendable partial answers are the sections of a sub-presheaf of Example 3 carved out by T: call it *the presheaf of local sections of* T. Its restriction maps are still just "forget the dropped variables" (a partial answer extendable to T stays extendable after forgetting variables).

The bare presheaf of Example 3 — all partial assignments, no constraint — is a sheaf for boring reasons, exactly like temperature maps. The interesting object is the constrained presheaf of local sections, and the interesting question is Section 4's question aimed at it: *does the constrained presheaf satisfy the gluing axiom over the given cover?* A constraint, like boundedness, may impose hidden global commitments that no window can see.

6 Local agreement versus a global solution

Two definitions carry the whole idea; they are the sheaf axioms of Section 4, specialized.

Compatible family (local agreement). Pick one local section s_j on each context U_j, such that any two agree on the variables their contexts share:

$$\text{for all } j,k \text{ and every } v \in U_j \cap U_k, \qquad s_j(v) = s_k(v).$$

This is the hypothesis of the gluing axiom: "everything is locally fine and the overlaps match." It is the cheap, checkable condition. (In cohomology this overlap-agreement is called the *cocycle condition.*)

Gluing (a global solution). The family *glues* if there is a single global configuration $f \in T$ whose restriction to each window U_j is exactly s_j — the conclusion of the gluing axiom. One answer that explains all the local pieces simultaneously. This is the expensive thing we actually want.

The obstruction. Every genuine global solution produces a compatible family (restrict it to each window). The question is the converse:

> *Does every compatible family glue?*

When the answer is *yes* for a constraint T and a cover, we say T **glues over that cover** — its presheaf of local sections behaves like a sheaf for that cover, and its *Čech obstruction vanishes* ($H^1 = 0$, in the language of the moral above). Local agreement then *is* global solvability. When the answer is *no*, some compatible family is locally perfect but globally impossible — the constraint has the bounded-functions disease — and the number of such stuck families is the size of the obstruction, a nonzero H^1 class.

7 A dictionary

STE / plain word	**Sheaf word**	**What it means**
Variable value space A_v	Stalk / local data	values a coordinate can take
Context U_j	Open set / patch	a window of variables
All partial answers on windows	Presheaf (of assignments)	data attached to regions
Forget the dropped variables	Restriction map	big-window data to small-window
Constraint / feasible set T	Sub-presheaf of local sections	the not-obviously-doomed pieces
Partial answer on U_j	Section over U_j	a value on that window
Extendable partial answer	Local section of T	a locally-valid piece
Overlaps match	Cocycle / gluing hypothesis	pieces agree where they share
One answer fits all pieces	Global section / gluing	a genuine global solution
Local determines global	Locality (separation) axiom	no invisible global data
Local agreement $\Rightarrow$ global	Gluing axiom / $H^1 = 0$	the obstruction vanishes
Stuck locally-fine families	Nonzero H^1 class	the coupling obstruction

8 One example, by hand

Take two variables x, y, each Boolean ($A_x = A_y = \{0, 1\}$), and the cover by the two single-variable windows $\{x\}$ and $\{y\}$. There is no overlap, so "overlaps match" is automatic — *every* combination of a local value for x and a local value for y is a compatible family. There are $2 \times 2 = 4$ of them.

A gluable constraint (uncoupled). Let T be the "rectangle" $\{0, 1\} \times \{0, 1\}$: anything goes. All four compatible families glue — each is just a global configuration read locally. Obstruction: zero. More generally any *product* constraint $\prod_v P_v$ (each variable independently confined to some allowed set P_v, no cross-variable link) glues over *every* cover. Product constraints are the continuous functions of this story: the property each window certifies is genuinely local.

A non-gluable constraint (coupled). Let T be the "diagonal" $\{(0, 0), (1, 1)\}$: the two variables are forced equal (the XOR/equality link). Locally each variable can still be 0 or 1 — both values extend to *some* element of T — so all four compatible families are still there. But only *two* of them, $(0, 0)$ and $(1, 1)$, come from a global configuration in T. The mixed family "$x \mapsto 0$, $y \mapsto 1$" is locally flawless and globally impossible: it is the $f(x) = x$ of bounded functions, a compatible family with no global section above it. Two stuck families remain: the obstruction is $2 \neq 0$. The coupling between x and y is invisible in any single window and shows up *only* as this obstruction.

That contrast — rectangle glues, diagonal does not — is the entire mechanism in miniature.

9 The dichotomy, and why it is useful

The example is not a coincidence. In the project's Lean library the two sides are theorems:

- **Uncoupled $\Rightarrow$ gluable.** Every product ("rectangular") constraint glues over every cover (`rectangular_cechVanishesCover`, module `Ste.CechCover`).
- **Gluable $\Rightarrow$ uncoupled.** If the obstruction vanishes, the constraint *must* be a rectangle — the product of its own per-variable projections (`rectangular_of_cechVanishes`, module `Ste.CechObstruction`).

- **The witness.** The diagonal's obstruction is computed to be exactly 2, a genuinely nonzero class (`diagonal_cechObstruction`, module `Ste.CechObstruction`).

So, in this setting, *gluable = rectangular = uncoupled* are one and the same, and the obstruction is a sharp detector of coupling.

Why it matters for solving problems:

1. **Gluable problems are cheap.** If the obstruction vanishes, you never assemble a global solution by search. You solve each window — in the extreme, each single variable — and local agreement is a *theorem* that the global answer exists. For the "frame" corpora the project studies (each source asserts unary `variable = value` facts), feasibility collapses to a one-variable-at-a-time scan: a value is consistent iff no two sources put different values on the same variable. This is the same fact as "pairwise agreement forces global agreement" — the Helly-number-2 property (`frame_hasHelly_two`, module `Ste.HellyConsistency`) — and it is why the project's solver certifies large corpora by a linear sweep instead of an all-subsets search.

2. **The obstruction localizes the hard part.** When it does *not* vanish, its support tells you *which* variables are genuinely entangled and must be reasoned about jointly. You pay the global price only there. Substitution-cipher key spaces (Carlson's setting, 3) are the coupled extreme: they are not rectangular, so the obstruction is essential and local reasoning cannot finish the job.

In one line: **the Čech obstruction is a computable measure of how far an STE problem is from being solvable window-by-window**, zero exactly when it factors into independent pieces.

10 A companion structure: the readings form a metric space

The gluing story above lives on the *constraint* side. There is a companion object on the *answer* side worth naming, because it turns out to carry more than an order — it carries a genuine *distance.*

Group the possible answers by what they identify: a "reading" is a partition of the underlying items into those forced to agree. One reading is *finer* than another if it splits at least as much, and this refinement relation makes the readings into a lattice (a meet/join order). This is exactly Shannon's 1953 *lattice of information*: a random variable, viewed by which distinctions it records, and ordered by "is a coarsening of." Its recent synthetic treatment [5] is what prompted this section.

The point Shannon already saw is that this lattice is *metric.* Assign each reading a *rank* — for a finite item set, the number of items minus the number of groups, i.e. how many merges the reading commits to. Then

$$D(P,Q) \;=\; \big(\mathrm{rank}(P\sqcup Q)-\mathrm{rank}\,P\big)+\big(\mathrm{rank}(P\sqcup Q)-\mathrm{rank}\,Q\big)$$

— the counting shadow of Shannon's entropic distance $H(X\mid Y)+H(Y\mid X)$ — is a true distance: it is zero exactly when P and Q are the same reading, it is symmetric, and it satisfies the triangle inequality. Crucially, the triangle inequality is not a separate miracle: it falls straight out of the same *submodularity* of the rank that governs the coupling story (larger overlaps cost less), so the metric and the obstruction spring from one structural fact. All of this is machine-checked (module `Ste.InfoDistance`); the entropy-valued distance itself — with genuine logs and probabilities — is the content of Delsol et al. [5] and is cited, not reproved.

So the feasible structure is not merely a set or an order. It is a metric space, and the same submodular rank that makes the Čech obstruction vanish for uncoupled problems is what makes the reading lattice metric. "Lighter than a metric space" was the wrong worry: the metric was there all along.

11 What is actually machine-checked

To keep the bright line the project insists on, here is what is a theorem (checked by Lean's kernel, sorry-free) versus what is intuition.

Theorems (names are Lean identifiers on the project's main branch):

- the compatible-family / gluing / vanishing definitions for any cover: `CompatibleFamily`, `GluesCover`, `CechVanishesCover` in `Ste.CechCover`;
- rectangles glue over every cover: `rectangular_cechVanishesCover` in `Ste.CechCover`;
- vanishing forces rectangularity (`rectangular_of_cechVanishes`) and the diagonal's obstruction is 2 (`diagonal_cechObstruction`), both in `Ste.CechObstruction`;
- the support-cover form — vanishing is equivalent to the constraint being generated by pieces whose scopes fit the cover: `cechVanishesCover_iff_exists_scopedGeneration` in `Ste.SupportCover`;
- the frame Helly-2 collapse: `frame_hasHelly_two` in `Ste.HellyConsistency`;
- the reading lattice is a metric space under the counting Shannon distance: `shannonDist_self`, `shannonDist_comm`, `shannonDist_eq_zero_iff` (separation), and `shannonDist_triangle` (triangle inequality, from rank submodularity), all in `Ste.InfoDistance`.

Intuition / framing, not claimed as proven here: the everyday presheaf and sheaf examples of Sections 3–4 (temperature, continuous functions, bounded functions) are standard textbook material, used for intuition only. And the word "cohomology" as the *full* Čech H^1 machinery — a cochain complex with a coboundary map and $H^1 = \ker / \operatorname{im}$ over a coefficient module — is the Abramsky and Brandenburger [1], Abramsky et al. [2] sheaf-theoretic framework the project draws on. This note uses the $H^1 = 0$ / obstruction-vanishing *level*, which is what is mechanized; the graded cochain-complex construction in full generality is cited, not reproved here.

The recast, restated

STE asks whether many sources share a common answer. A constraint induces a presheaf: on each window, the partial answers that could still extend to a full solution, with restriction maps that just forget variables. "Every window is fine and the overlaps match" is the hypothesis of the sheaf gluing axiom, and is cheap to check. Whether that local agreement assembles into one global answer is exactly the question of whether this presheaf glues — and the failure, when it fails, is a single computable obstruction. It vanishes exactly for the uncoupled (rectangular) problems, and then STE is a local computation; it is nonzero exactly when the problem is genuinely coupled, and then it measures and locates the coupling, as boundedness haunts the bounded functions. That obstruction is the cohomology, and that equivalence is the recast.

References

[1] Samson Abramsky and Adam Brandenburger. The sheaf-theoretic structure of non-locality and contextuality. *New Journal of Physics*, 13:113036, 2011. doi: 10.1088/1367-2630/13/11/113036. arXiv:1102.0264.

[2] Samson Abramsky, Shane Mansfield, and Rui Soares Barbosa. The cohomology of non-locality and contextuality. *Electronic Proceedings in Theoretical Computer Science*, 95:1–14, 2012. doi: 10.4204/EPTCS.95.1. Proc. 8th International Workshop on Quantum Physics and Logic (QPL 2011). arXiv:1111.3620.

[3] Albert H. Carlson. *Set Theoretic Estimation Applied to the Information Content of Ciphers and Decryption*. Ph.d. dissertation, University of Idaho, Moscow, Idaho, USA, May 2012. Retrieved via Internet Archive snapshot (2024-07-13) of the author's ResearchGate full text, publication 349064680.

[4] Patrick L. Combettes. The foundations of set theoretic estimation. *Proceedings of the IEEE*, 81 (2):182–208, 1993. doi: 10.1109/5.214546. Retrieved from the author's page, `https://pcombet.math.ncsu.edu/proc.pdf`.

[5] Idris Delsol, Olivier Rioul, Julien Béguinot, Victor Rabiet, and Antoine Souloumiac. An information theoretic condition for perfect reconstruction. *Entropy*, 26(1):86, 2024. doi: 10.3390/e26010086. PMC10814784. Shannon's 1953 lattice of information with the Shannon and Rajski entropic metrics.